Consensus committee

A consensus committee was established, comprising expert representatives from each medical specialty involved in the management of Incontinentia Pigmenti (IP). All experts are co-authors of the manuscript. Multiple meetings—conducted in person and via conference calls—were held until full agreement was reached on the consensus recommendations.

Results

A review of the literature revealed a paucity of randomized controlled trials, with most studies involving small, heterogeneous patient groups and lacking standardized outcome measures. The majority of evidence was classified as level 4 (retrospective studies, case series) and grade C (low scientific evidence). Over 15 consensus committee meetings were conducted. The final manuscript, approved by all French and Italian experts from two ERN-Skin Healthcare Professional (HCP) networks, provides the following consensus-based recommendations, reflecting expert professional agreement for the management of this rare disease.

Diagnosis

In most cases, the diagnosis of IP is primarily clinical in neonates and infants, and can be confirmed through molecular genetic testing.

Clinical diagnosis of IP

The clinical diagnostic criteria for IP, originally proposed by Landy and Donnai in 1993, have been updated to include recent clinical and histological findings. This updated version is summarized in Table 1.

In the absence of a family history, the presence of at least one major criterion is sufficient to establish the diagnosis of IP. If a first-degree female relative is affected, the presence of one minor criterion is sufficient for diagnosis. The complete absence of minor criteria should raise doubts about the diagnosis.

Table 1: Updated diagnostic criteria for IP

Major criteria - Typical neonatal rash (erythema and vesicles) along Blaschko's lines (Stage 1)
- Verrucous papules or plaques along Blaschko's lines (Stage 2)
- Typical hyperpigmentation along Blaschko's lines, fading in adolescence (+++) (Stage 3)
- Linear, atrophic, hairless skin lesions on limbs (Stage 4) or scarring alopecia of the scalp vertex (Stages 3 or 4)
- Dental abnormalities: hypodontia/oligodontia, peg-shaped or conical teeth, altered molar cusp patterns, delayed tooth eruption
- Common recurrent rearrangement: deletion of exons 4–10 in the *IKBKG* gene

Minor criteria - Eosinophilia (typically in Stage 1)
- Hair abnormalities: alopecia or woolly hair (dull and dry)
- Nail changes: punctuate depressions, onychogryphosis (ram's horn nails)
- Mammary gland involvement (hypoplasia, asymmetry, hypogalactia) and/or nipple abnormalities (inverted, supernumerary nipples, feeding difficulties)
- Characteristic skin histology (e.g., eosinophilic spongiosis)
- Ocular findings: peripheral retinal neovascularization

Clinical diagnosis: The importance of the neonatal period

Cutaneous lesions (see Fig. 1) are key diagnostic features and constitute major criteria. Four distinct stages are well recognized and are summarized in Table 2, along with their main differential diagnoses.

The initial inflammatory stage (Stage 1) presents as vesiculopustular lesions, typically followed by papular-verrucous plaques (Stage 2), then hyperpigmented macules along Blaschko's lines (Stage 3). The lesions may be widespread or limited in extent.

These skin manifestations are highly suggestive of IP when they: - Occur in a female neonate
- Show a linear distribution following Blaschko's lines, often with acral predominance
- Resolve spontaneously in early childhood (Stages 1 and 2) or gradually fade during adolescence (Stage 3)